# Daniel Alejandro Silva Rojas

## Senior Backend Engineer | Full-Stack | Node.js · TypeScript · Python · Rust | Real-Time Systems

Caracas, Venezuela (UTC−4) · Remote · +58 414-231-7561 · dsrglrm@gmail.com
danielsilvarojas.dev · github.com/Dan178A · linkedin.com/in/daniel-alejandro-silva-rojas

Available immediately for full-time remote roles. Hireable as an independent contractor (W-8BEN) or through an Employer of Record such as Deel.

## SUMMARY

Senior Backend Engineer with 5+ years building and operating production systems end to end. I design event-driven, low-latency services (WebSockets, REST, gRPC) in Node.js/TypeScript, Python and Rust, model data in PostgreSQL and MySQL, and ship on AWS with Docker and CI/CD. At Ea2technology (Canada) I replaced HTTP polling with a full-duplex WebSocket architecture that cut latency from 200–500 ms to 1–10 ms and bandwidth by up to 95%, and led the migration from legacy monoliths to cloud-native microservices. I also deliver the frontend when the team needs it (Vue 3, Nuxt 3, React), and I built and run a production e-commerce platform on Nuxt 3 and PostgreSQL. English B2 (professional), five years of daily work with a Canada-based team.

## SKILLS

**Languages:** TypeScript, JavaScript, Python, Rust, C++, SQL

**Backend:** Node.js, NestJS, Express, Nitro, FastAPI, Axum, Tokio, REST APIs, gRPC, WebSockets, Socket.IO, TCP sockets, Microservices, Event-Driven Architecture

**Data:** PostgreSQL (schema design, indexing, query optimization, RLS), MySQL, MariaDB, Redis, MongoDB, Supabase, Polars, pandas, NumPy, SciPy

**Cloud & DevOps:** AWS (EC2, S3, RDS, Lambda, CodePipeline), Docker, CI/CD (GitHub Actions, GitLab CI), Vercel, Linux

**Frontend:** Vue 3, Nuxt 3 (SSR), React, Tailwind CSS, Three.js, D3, PWA

**Real-time & AI:** Low-latency systems, concurrency, profiling, offline-first sync, industrial IoT telemetry, RAG, speech-to-text / TTS pipelines, OpenCV, PyTorch

## EXPERIENCE

**Lead Performance & Systems Engineer — Ea2technology**
*Remote (Canada)* | *Jul 2021 – Sep 2026*

- Designed a full-duplex WebSocket architecture that replaced HTTP polling for real-time dashboards: latency from 200–500 ms to 1–10 ms and up to 95% less bandwidth.
- Built an industrial Digital Twin platform ingesting high-frequency sensor telemetry for real-time structural monitoring of refinery equipment, from ingestion through the API layer to the operator UI.
- Led the migration of legacy monoliths to cloud-native microservices on AWS, defining service boundaries, deployment flow and architectural trade-offs.
- Created and maintained the main industrial web platform (Nuxt/Vue, Express/TypeScript, Sequelize/MySQL, AWS): ~2,000 commits, integration point for a ~14-person team, owner of the build and release cycle.
- Built an offline-first desktop ecosystem with bidirectional cloud↔local sync (Python orchestrator, Node/TypeScript services, FastAPI, MariaDB, S3) so plant teams work with no internet and reconcile automatically.
- Built a desktop app for coordinated control of networked industrial IoT hardware (Python, PyQt5, Socket.IO), designed to run unattended on unreliable networks; in field use for 5 years.
- Optimized compute-intensive data pipelines in C++ and Rust for memory safety and concurrency under sustained load; integrated an AI audio transcription and synthesis pipeline.

**Senior Software Engineer (Contract) — Vessel Inspection and Assessment Corp (VIAAC)**
*Remote, maritime / oil & gas | Jan 2023 – Jul 2025*

- Built and containerized backend services with NestJS and Docker; set up CI/CD with GitLab CI, GitHub Actions and AWS CodePipeline (CodeCommit, CodeArtifact, CodeBuild, CodeDeploy).
- Audited 4 production stacks and fixed 3 memory leaks and concurrency bottlenecks in point-cloud processing (up to 1.2M points): a sequential job dropped from 12–14 hours to 5 hours and data-navigation response from ~1 s to 15–20 ms.
- Designed TCP socket and WebSocket services with custom file-transfer protocols for high-frequency environments.
- Developed video stabilization engines (optical flow, matrix interpolation; Python, NumPy, SciPy, OpenCV) with 65–80% jitter reduction at 60 fps, HD to Full HD.
- Presented audit findings and remediation standards to the client's engineering team and specified the changes they adopted.

**IT Support Specialist — RenéDessés de Venezuela**
*Caracas, Venezuela | Jan 2021 – Sep 2021*

- Sole systems administrator for an 80-person company: Cisco network, MySQL Server, Apache/Linux servers and user support.

## PROJECTS

**DropAudio CCS — production e-commerce platform (dropaudioccs.com)**
*Nuxt 3 · Node.js (Nitro) · PostgreSQL · Supabase · Vercel*

- Built and run a live store end to end: API routes, PostgreSQL schema with Row Level Security, indexes and optimized queries, admin panel and storefront; 120+ verified customer reviews.
- Implemented a multi-currency checkout (USDT, Zinli, Pago Móvil) with live exchange rates, order state handling and scheduled jobs (pg_cron) for Web Push notifications.

**RealtimeVoiceAssistant — Rust (Tokio) · Python · Gemini Live**
Real-time conversational voice assistant with streaming transcription and speech synthesis on an async Rust runtime.

**extract_rif — FastAPI · PaddleOCR · Docker**
HTTP microservice that extracts structured data from tax documents (PDF or image) with a dual-engine OCR pipeline.

## EDUCATION

**B.Sc. in Computer Science** — Universidad del Zulia (LUZ-IUTA), Venezuela | Graduated Apr 2025 (completed while working full-time)

**Associate Degree (TSU) in Information Technology** — Universidad del Zulia (LUZ-IUTA) | 2018 – 2021

## LANGUAGES

Spanish (native) · English (B2, professional)